Face Verification & Identification

Neural Networks and Deep Learning - Spring 2026

Course Code: CIE 555

1. Overview

This assignment requires students to develop, evaluate, and compare advanced Deep Learning architectures for **Face Verification** using the **LFW (Labeled Faces in the Wild)** dataset. Students are expected to move beyond simple classification by building **Siamese Networks**. The task involves constructing custom convolutional backbones, applying transfer learning, and implementing multiple metric learning loss functions.

2. Learning Objectives

Upon successful completion of this assignment, students will be able to:

- Perform advanced data preprocessing and programmatic filtering on real-world datasets.
- Design and implement Siamese Network architectures for similarity learning.
- Manually implement and compare **Contrastive Loss** and **Triplet Loss** functions.
- Integrate pre-trained models as feature extractors within a Siamese framework.
- Produce rigorous quantitative evaluations of distance-based embeddings.

3. Dataset

The **Labeled Faces in the Wild (LFW)** dataset, a database of face photographs designed for studying the problem of unconstrained face recognition, is to be used.

- **Source:** https://www.kaggle.com/datasets/jessicali9530/lfw-dataset
- **Note:** Students must handle the raw folder structure and programmatically manage the distribution of images per person.

4. Mandatory Requirements

4.1 Data Handling & Filtering

Students must programmatically process the LFW directory. The pipeline should:

- Parse the folder structure to identify individuals and their corresponding images.

- Implement a filtering logic to select appropriate candidates for pair/triplet generation.
- Justify the selection criteria used to ensure effective model convergence.

4.2 Model Architecture

Two distinct architectural strategies must be implemented and compared:

- **Custom CNN Backbone:** A Siamese network built from first principles (Conv2D, Pooling, Dropout, BatchNormalization).
- **Transfer Learning Backbone:** A Siamese network utilizing a pre-trained backbone (e.g., MobileNetV2, ResNet50, or VGG16) as a feature extractor.

4.3 Loss Function Implementation

The models must be trained and evaluated using two different loss strategies:

- **Contrastive Loss:** Requires a pair-based data pipeline (Same/Different).
- **Triplet Loss:** Requires a triplet-based data pipeline (Anchor, Positive, Negative).
- *Note:* Both loss functions must be implemented manually (Custom Loss Functions).

4.4 Evaluation & Visualisation

The following outputs are required for all experiments:

- Training and validation loss/accuracy curves.
- **Distance Distribution Plot:** Visualizing the Euclidean distance between positive pairs versus negative pairs.
- **Inference Pipeline:** A function that takes two unseen images and outputs the distance, the similarity decision, and a visual display of the pair.

5. Grading Rubric (Total: 10 Marks)

Component	Criteria	Weight
Data Engineering	Successful parsing and programmatic filtering of the raw LFW dataset.	1.0
Architecture Design	Implementation of both Custom CNN and Transfer Learning Siamese networks.	3.0

Component	Criteria	Weight
Loss Implementation	Manual implementation and training of Contrastive and Triplet losses.	3.0
Evaluation	Quality of metrics, distance plots, and the inference function.	2.0
Technical Report	Discussion of challenges, experiments, and comparative analysis.	1.0

6. Deliverables

1. **Jupyter Notebook (.ipynb):** A fully annotated notebook containing all code, preprocessing, and experimental results.
2. **Technical Report (PDF):** A 2-page report summarizing the data strategy, model architecture choices, and a critical discussion of the results (Contrastive vs. Triplet).